USN | | | | | | | | | | [24MTEEM103]

# FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025

## ENERGY CONSERVATION AND MANAGEMENT

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

| | | | CO's | Blooms Level | Marks |
|---|---|---|---|---|---|
| **1.** | **a.** | Analyze the Indian energy scenario, focusing on sectorial energy consumption patterns in domestic, industrial, and other sectors. | 1 | 4 | **12** |
| | **b.** | Explain the concept of energy pricing and its impact on energy conservation efforts. | 1 | 2 | **08** |
| | | | | | |
| **2.** | **a.** | Analyze the energy conservation potential in industrial furnaces. What are the key areas for improvement? | 2 | 4 | **12** |
| | **b.** | Discuss the energy efficiency improvements possible in industrial compressor systems. | 2 | 3 | **08** |
| | | | | | |
| **3.** | **a.** | Discuss the importance of power factor correction. What methods can be employed to improve power factor in industrial settings? | 3 | 3 | **10** |
| | **b.** | Describe the process of load scheduling and shifting. How does it help in energy conservation? | 3 | 2 | **10** |
| | | **OR** | | | |
| **4.** | **a.** | Explain different types of motor drives used in industrial applications. How do they impact energy efficiency? | 3 | 2 | **14** |
| | **b.** | Summarize the key provisions of the Electricity Act that relate to energy conservation and efficiency. | 3 | 2 | **06** |
| | | | | | |
| **5.** | **a.** | Discuss the organizational structure desired for the smooth functioning of energy management activities in an industrial setting. | 5 | 2 | **10** |
| | **b.** | Explain how a CUSUM (Cumulative Sum of Differences) chart is drawn. Provide an example to illustrate its use in energy management. | 6 | 3 | **10** |
| | | **OR** | | | |
| **6.** | **a.** | Elucidate the factors contributing to the successful initiation and maintenance of an energy management program. | 5 | 4 | **12** |
| | **b.** | Define "benchmarking" in the context of energy management. List and explain its key benefits for an organization. | 6 | 2 | **08** |

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| **7.** | **a.** | Present a case study of energy conservation measures in a textile industry, focusing on steam systems. What were the main areas of improvement and their economic analysis? | 4 | 4 | **12** |
| | **b.** | Discuss a case study of implementing energy-efficient windows in a high-rise office building. | 4 | 4 | **08** |
| | | **OR** | | | |
| **8.** | **a.** | Analyze a case study of implementing variable speed drives in an industrial pumping system. How were the energy savings calculated, and what will be the payback period? | 4 | 4 | **12** |
| | **b.** | Present a case study of energy conservation in residential sector focusing on improving home insulation and its impact on energy consumption. | 4 | 4 | **08** |